Agentic Quadruped SLAM & Autonomous Exploration: Benchmark Results

Generated 2026-10-12 from a closed-loop CoppeliaSim run (robot_working.ttt) with no ground truth used at runtime. Every non-OURS value is quoted from the cited source.

Table 1. Our system: measured results of a full autonomous exploration run

Category	Metric	OURS
Exploration	Mission time / path walked	1800 s / 160.5 m
Exploration	Rooms cleared / segmented, doorways found	6 / 8, 12
Exploration	Free floor area mapped	63.6 m2
Localization	ATE RMSE, first 5.3 m (short-path benchmark)	10.0 cm
Localization	ATE RMSE, first 148 m (drift % of path)	30.3 cm (0.21 %)
Localization	Mean heading error / loop closures	1.11 deg / 120
Mapping	Wall accuracy / completeness / Chamfer	15.8 cm / 6.6 cm / 11.2 cm
Mapping	Wall F1 (+/-15 cm) / free-space IoU	76.5 % / 78.4 %
Efficiency	Tracking / mapping time per update (CPU only)	5.8 ms / 10.6 ms
Efficiency	Map memory / processor	0.37 MB / AMD Ryzen 7 7735HS with Radeon Graphics

Sensors: one forward 60 deg RGB-D camera, a gyro (simulated bias 0.05 deg/s + noise) and gait-command odometry. The run was not used for tuning (held-out). The robot ran 1800 s; at 1640 s an idle-gait creep against a wall (since fixed) moved it while blind, so accuracy is reported up to that point; whole-run ATE 84.7 cm (0.53 %).

Table 2. Localization on short quadruped paths (~5 m)

Method	Sensor (field of view)	Robot	Paths	Mean ATE RMSE	Drift (% of path)	Source
Cartographer	2D LiDAR, 360 deg + IMU	Unitree A1 (real)	4 paths, 5.0-5.3 m	63.3 cm	12.63 %	Chen & Dellaert
A1 SLAM (factor graph)	2D LiDAR, 360 deg + IMU	Unitree A1 (real)	4 paths, 5.0-5.3 m	8.8 cm	1.74 %	Chen & Dellaert
OURS	**RGB-D camera, 60 deg + gyro**	**Quadruped (sim)**	**1 path, 5.3 m**	**10.0 cm**	**1.89 %**	**This work**

Published values: mean over the four trajectories of Chen & Dellaert (Tables 1-2, motion-capture ground truth). With one-sixth of the field of view, our error is within 1.2 cm of A1 SLAM and 6.3x lower than Cartographer. Two of their four paths were deliberately aggressive (0.22-0.34 m/s); our robot averaged 0.09 m/s.

Table 3. Long-range drift on quadrupeds: drift as % of distance travelled

Method	Sensors	Drift (% of path)	ATE RMSE	Source
OURS	**RGB-D 60 deg + gyro (single camera)**	**0.21 %**	**0.303 m**	**This work**
DPV-SLAM	camera (Sevensense, global shutter)	0.41 %	0.967 m	Corlito et al. 2026
RTAB-Map	RGB-D (ZED2i)	1.03 %	2.413 m	Corlito et al. 2026
ORB-SLAM3	RGB-D (ZED2i)	3.50 %	8.214 m	Corlito et al. 2026
FAST-LIVO2	LiDAR + camera + IMU (Honeywell)	0.04 %	0.105 m	Corlito et al. 2026
ORB-SLAM3	stereo-inertial (ADIS IMU)	0.10 %	0.229 m	Corlito et al. 2026
ORB-SLAM3	stereo (global shutter)	0.14 %	0.330 m	Corlito et al. 2026
RTAB-Map	stereo (global shutter)	0.25 %	0.579 m	Corlito et al. 2026

Published values: ANYmal D, GrandTour dataset, 7 missions of 145-332 m (Corlito et al. 2026, Tables II-V; drift = mean ATE / mean mission length, derived by us). Ours: 148 m in simulation. Among camera-only systems (no stereo or LiDAR) ours drifts least (5.0x lower than the best RGB-D result); stereo and LiDAR-inertial systems remain ahead.

Table 4. Computational footprint

System	Compute	Tracking time / frame	Memory	Source
OURS	**CPU only (AMD Ryzen 7 7735HS with Radeon Graphics)**	**5.8 ms**	**0.37 MB (2D grid + rooms)**	**This work**
ESLAM	not stated	68.5 ms	20.73 MB (neural scene model)	Deng et al. 2025
Co-SLAM	not stated	54.8 ms	26.39 MB (neural scene model)	Deng et al. 2025
MNE-SLAM	not stated	22.1 ms	24.45 MB (neural scene model)	Deng et al. 2025
ORB-SLAM2 (RGB-D)	Intel Core i7-4790	25.6 ms	-	Mur-Artal & Tardos
FAST-LIVO2 (LiDAR + camera + IMU)	125 % CPU load	-	4004 MB process RAM	Corlito et al. 2026
ORB-SLAM3 (stereo)	290 % CPU load	-	1188 MB process RAM	Corlito et al. 2026
RTAB-Map (stereo)	116 % CPU load	-	768 MB process RAM	Corlito et al. 2026
DPV-SLAM (camera)	82 % CPU load	-	4516 MB process RAM	Corlito et al. 2026

Different hardware per row; read as orders of magnitude. Our map is 56x smaller than the smallest neural scene model and needs no GPU, which fits the project's Raspberry Pi target.

Table 5. Capabilities compared (as documented by each source)

System	Platform / sensing	Explores an unknown building autonomously	Room + doorway graph	Semantic knowledge graph	Agentic / multi-agent design	CPU only
OURS	**Quadruped, 60 deg RGB-D**	**Yes**	**Yes (rooms + doorways)**	**Yes (rooms, doorways, objects)**	**7 cooperating software agents**	**Yes**
Boston Dynamics Spot	Quadruped, 360 deg sensing	No (records route first)	-	-	-	-
ANYbotics ANYmal	Quadruped, 360 deg LiDAR + 6 depth	Not stated (inspection routes)	-	-	-	-
Unitree Go2	Quadruped, 4D LiDAR	No (user draws path)	-	-	-	-
A1 SLAM	Quadruped, 2D LiDAR	No (SLAM only)	No	No	No	not stated
FARE (LLM + RL)	Wheeled, 16-ch 3D LiDAR	Yes	Topological graph	LLM over text description	LLM + RL policy	not stated
MNE-SLAM	RGB-D datasets	No (SLAM only)	No	No	Multiple robots	not stated

Sources: manufacturer pages and Spot SDK docs (read 2026-10-11), A1 SLAM, FARE and MNE-SLAM papers. '-' = not applicable or not documented. Our combination - prior-free exploration of an unknown building by a legged robot with a single camera, room/doorway decomposition and a semantic graph, all on a CPU - is not offered by any listed system.

Table 6. Ablation: contribution of each component (identical recorded data)

Configuration	ATE RMSE	Drift	Heading error	Loop closures
OURS full system	**8.5 cm**	**0.22 %**	**0.73 deg**	**28**
without ICP (gait odometry + gyro only)	61.2 cm	1.60 %	10.10 deg	0
without loop closure	31.7 cm	0.83 %	0.70 deg	0
without degeneracy filter	15.9 cm	0.42 %	0.73 deg	27
without Manhattan heading prior	15.3 cm	0.40 %	2.23 deg	28
with keyframes 1.5 m / 40 deg (previous default)	14.4 cm	0.38 %	0.76 deg	10

Replay of recordings/arena_gtnav_420s.npz (420 s, 38.2 m, recorded with ground-truth navigation) through the SLAM agent. Novel components: gyro-fixed ICP with degeneracy filtering, Manhattan-world heading prior, and dense keyframe loop closure.

Where we are behind

- Long-range drift: LiDAR-inertial SLAM (FAST-LIVO2) drifts 0.04 % vs our 0.21 %, and stereo systems are also ahead (Table 3).
- Evidence base: one simulated run; published systems report real sensors and several sequences, and we have no result on a public benchmark (e.g. TUM RGB-D) yet.
- Exploration: 6 of 8 rooms cleared in 30 min at 0.09 m/s; 64 frontiers left unexplored.
- Odometry: gait-command odometry under-counted the distance by 19 % (foot slip and turning are not modelled).
- Robustness: a blind spot at very close range (camera minimum range) let the robot move unobserved; the heading prior assumes walls meet at right angles.

What to improve to be the best in this category

- Estimate foot slip online (leg-velocity bias state, as in VILENS) instead of a fixed gait-speed curve.
- Add close-range sensing (a second depth camera or low-cost 2D LiDAR / ToF bumpers) to remove the blind zone and widen the 60 deg field of view.
- Match scans to a local map instead of the previous scan and use a real pose-graph optimiser (GTSAM / g2o) to bring drift towards stereo-level results.
- Make exploration faster and complete: higher gait speed, semantic frontier scoring (VLFM) and LLM room ordering from the knowledge graph (FARE / SayPlan); then compare against explore_lite / TARE in the same arena.
- Strengthen the paper evidence: 5+ repeated runs (mean +/- std), a real-robot or TUM RGB-D result, and timing on the Raspberry Pi target.

References

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